Name

$M\sharp$

Models

ODE $y' = A\,y$

If A is diagonalizable with LI eigenvectors $v_1, \ldots, v_n$ and $A\,v_i = \lambda_i\,v_i$ then

- 3.4 $y(t) = \sum_{i=1}^{n} c_i\,e^{\lambda_i t}\,v_i$ is the solution of $y' = A\,y$ with $y(0) = y_0$ where
  $V\,c = [v_1 \mid \ldots \mid v_n]\,c = y_0$.
- 3.5 Even true for complex evals $\lambda = a \pm b\,i$ and evecs $v = p \pm q\,i$
  - But every computation involves complex numbers
  - Taylor Series show that
    - $e^{(a \pm b\,i)\,t} = e^{(a\,t)}\,e^{(\pm i\,b\,t)} = e^{a\,t}(\cos(b\,t) \pm i\sin(b\,t))$
  - Direct substitution gives
    - $y_+ = e^{(a+b\,i)\,t}(p + q\,i) = e^{a\,t}(\cos(b\,t)\,p - \sin(b\,t)\,q) + i\,e^{a\,t}(\cos(b\,t)\,q + \sin(b\,t)\,p)$
    - $y_- = e^{(a-b\,i)\,t}(p - q\,i) = e^{a\,t}(\cos(b\,t)\,p - \sin(b\,t)\,q) - i\,e^{a\,t}(\cos(b\,t)\,q + \sin(b\,t)\,p)$
  - We need two LI solutions. Linearity gives the real solutions
    - $\frac{1}{2}(y_+ + y_-) = e^{a\,t}(\cos(b\,t)\,p - \sin(b\,t)\,q)$
    - $\frac{1}{2\,i}(y_+ - y_-) = e^{a\,t}(\cos(b\,t)\,q + \sin(b\,t)\,p)$
- 3.5 Missing Eigenvector. Theorem 3.5.1
  - An eigenvalue λ_1 that is a quadratic factor $(\lambda - \lambda_1)^2$ in the characteristic polynomial of A has an eigenvector v_1 satisfying $A\,v_1 = \lambda_1\,v_1$ and <u>should</u> have a second LI eigenvector.
  - Eigenvectors v give solutions $e^{\lambda_1 t}\,v$.
    - If there is no second eigenvector we can find a second LI solution of the form
      $t\,e^{\lambda_1 t}\,v + e^{\lambda_1 t}\,u$
      where u satisfies
      $(A - \lambda_1\,I)\,u = v$

3.6 ODE $y' = A\,y + b$

If $y_c(t) = \sum_{i=1}^{n} c_i\,e^{\lambda_i t}\,v_i$ is an n parameter solution of $y' = A\,y$ and $y_p' = A\,y_p + b$ then

$y = y_c(t) + y_p(t)$

is a n parameter solution of $y' = A\,y + b$

- $y = y_c + y_p$ is the general solution
  - y_c is called the complementary solution
  - y_p is called a particular solution
- You can frequently guess the form of a particular solution y_p. This is called undetermined coefficients.
  - If b is a polynomial you guess a polynomial.
  - If b contains exponentials you guess matching exponentials.
  - If b contains sines or cosines you guess matching sines and cosines.

3.7 Fundamental Matrix and Variation of Parameters

The complementary solution $y_c(t) = \sum_{i=1}^{n} c_i e^{\lambda_i t} v_i$ of $y' = A\, y$ has the form

$$y_c = \left[e^{\lambda_1 t} v_1 \mid e^{\lambda_2 t} v_2 \mid \ldots \mid e^{\lambda_n t} v_n \right] \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} = \Phi(t)\, c$$

The matrix Φ formed by stacking n linearly independent solutions of an ODE $y' = A\, y$ as columns is the fundamental matrix of the ODE.

- Since each column satisfies $y' = A\, y$ the matrix Φ satisfies the matrix ODE $\Phi' = A\, \Phi$
- Plugging $y(t) = \Phi(t)\, u(t)$ into $y' = A\, y + b$ gives
  - $\Phi'(t)\, u(t) + \Phi(t)\, u'(t) = A\, \Phi(t)\, u(t) + b$
  - $A\, \Phi(t)\, u(t) + \Phi(t)\, u'(t) = A\, \Phi(t)\, u(t) + b$
  - $\Phi(t)\, u'(t) = b \Longrightarrow u(t) = \int^t \Phi(\tau)^{-1}\, b(\tau)\, d\tau + c$
- A general solution formula!
  - $y(t) = \Phi(t) \left(\int^t \Phi(\tau)^{-1}\, b(\tau)\, d\tau + c \right) = \underbrace{\Phi(t) \left(\int^t \Phi(\tau)^{-1}\, b(\tau)\, d\tau \right)}_{\longleftarrow\ y_p\ \longrightarrow} + \underbrace{\Phi(t)\, c}_{\leftarrow\ y_c\ \rightarrow}$
  - This process is called variation of parameters